Worked Answer Key: Practice Problems, AP Precalc Unit 1

Unit 1: Polynomial and rational functions

Rates of change

W1. Average rate of change of $f(x) = 3x - 7$ on $[2, 6]$.

- $f(6) = 18 - 7 = 11$ and $f(2) = 6 - 7 = -1$.
- $\dfrac{f(6) - f(2)}{6 - 2} = \dfrac{11 - (-1)}{4} = \dfrac{12}{4} = \boxed{3}$
- Check: a line's average rate of change is always its slope, and the slope here is 3.

W2. Average rate of change of $f(x) = x^2$ on $[0, 2]$ and $[2, 4]$.

- $[0, 2]$: $\dfrac{f(2) - f(0)}{2 - 0} = \dfrac{4 - 0}{2} = 2$
- $[2, 4]$: $\dfrac{f(4) - f(2)}{4 - 2} = \dfrac{16 - 4}{2} = 6$
- The second is bigger. The rate of change is increasing, so the graph gets steeper as x grows: **concave up**.

1. Table: $f = 1, 4, 9, 16, 25$ at $x = 0, \ldots, 4$.

- (a) Subtract consecutive outputs: $4 - 1 = 3,\ 9 - 4 = 5,\ 16 - 9 = 7,\ 25 - 16 = 9$. The changes are **3, 5, 7, 9**.
- (b) The changes are getting bigger, so the rate of change is **increasing** and f is **concave up**.
- (c) Take differences of the differences: $5 - 3 = 2$, $7 - 5 = 2$, $9 - 7 = 2$. The second differences are constant, so f is **quadratic**. (It's $(x + 1)^2$.)

2. $f(x) = -x^2 + 6x$ on $[0, 2]$, $[2, 4]$, $[4, 6]$.

- Values: $f(0) = 0,\ f(2) = -4 + 12 = 8,\ f(4) = -16 + 24 = 8,\ f(6) = -36 + 36 = 0$.
- $[0, 2]$: $\frac{8-0}{2} = 4$. $[2, 4]$: $\frac{8-8}{2} = 0$. $[4, 6]$: $\frac{0-8}{2} = -4$.
- The rates **4, 0, −4** keep decreasing, so f is **concave down**. It rises, levels off at the top, then falls.

3. $h(t) = -16t^2 + 64t$ on $[1, 3]$.

- $h(1) = -16 + 64 = 48$ and $h(3) = -144 + 192 = 48$.
- $\dfrac{48 - 48}{3 - 1} = \boxed{0 \text{ ft/s}}$
- **Meaning:** the ball is at the same height (48 ft) at $t = 1$ and $t = 3$, so its net change in height over that interval is zero.
- **Not the meaning:** the ball didn't sit still. It went up to its peak (at $t = 2$) and came back down. An average hides what happens in between.

4. Increasing and concave down.

- Increasing means the rate of change is **positive**.
- Concave down means the rate of change is **decreasing**.
- Together: the function is still going up, but more slowly over time.
- Examples: a plant's height as it reaches full size; a car's speed as it approaches the speed limit; a learning curve.

Polynomials

W3. Zeros of $f(x) = (x-2)(x+5)$.

- A product is 0 when a factor is 0: $x-2=0$ or $x+5=0$.
- $\boxed{x=2,\ x=-5}$

W4. $f(x) = -2x^3 + x - 4$.

- Degree: highest power, **3**. Leading coefficient: **−2**.
- Odd degree with a negative leading coefficient: **up on the left, down on the right**.
- In limits: $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = -\infty$.

5. $f(x) = (x+2)(x-1)^2(x-3)$.

- (a) Set each factor to 0:

Zero	Factor	Multiplicity	Behavior
-2	$(x+2)^1$	1 (odd)	crosses
1	$(x-1)^2$	2 (even)	bounces
3	$(x-3)^1$	1 (odd)	crosses

- (b) Degree = sum of multiplicities $= 1+2+1 = 4$. Leading term: $x \cdot x^2 \cdot x = x^4$, so the leading coefficient is $+1$. Even degree, positive: **up on both ends**.
- (c) y-intercept: plug in $x=0$: $(2)(-1)^2(-3) = (2)(1)(-3) = \boxed{-6}$.

6. All zeros of $x^3 - x^2 + 4x - 4$.

- Group: $(x^3 - x^2) + (4x - 4) = x^2(x-1) + 4(x-1)$.
- Factor out $(x-1)$: $(x-1)(x^2+4)$.
- $x-1=0$ gives $x=1$.
- $x^2+4=0$ gives $x^2=-4$, so $x = \pm\sqrt{-4} = \pm 2i$.
- Zeros: $\boxed{1,\ 2i,\ -2i}$ (3 zeros for degree 3 ✓).

7. Degree 3, real coefficients, zeros 2 and $1+i$.

- Real coefficients mean complex zeros come in conjugate pairs, so $1-i$ is the third zero.
- Pair the complex factors first: $(x-(1+i))(x-(1-i))$. Group as $((x-1)-i)((x-1)+i)$, a difference of squares: $(x-1)^2 - i^2 = x^2 - 2x + 1 + 1 = x^2 - 2x + 2$.
- Multiply by $(x-2)$: $(x-2)(x^2-2x+2) = x^3 - 2x^2 + 2x - 2x^2 + 4x - 4$.
- Combine: $\boxed{x^3 - 4x^2 + 6x - 4}$

8. Even, odd or neither? Plug in $-x$.

- $f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$: **even**.
- $g(-x) = (-x)^3 + (-x) = -x^3 - x = -(x^3 + x) = -g(x)$: **odd**.
- $h(-x) = -x^3 + 1$. That's not $h(x) = x^3 + 1$, and not $-h(x) = -x^3 - 1$: **neither**.

9. End behavior of $p(x) = -3x^5 + 2x^2 - 1$.

- Leading term $-3x^5$: odd degree, negative coefficient, so up on the left and down on the right.
- $\lim_{x\to\infty} p(x) = -\infty$ and $\lim_{x\to-\infty} p(x) = \infty$

Rational functions

W5. $f(x) = \dfrac{1}{x-3}$.

- Vertical asymptote: the denominator is 0 at $x = 3$ and the top isn't, so $x = 3$.
- Horizontal asymptote: top degree 0, bottom degree 1. Bottom bigger, so $y = 0$.

10. $r(x) = \dfrac{2x^2 - 8}{x^2 - x - 6}$.

- Factor: top $2(x^2 - 4) = 2(x-2)(x+2)$; bottom $(x-3)(x+2)$.
- $(x+2)$ cancels, giving a **hole** at $x = -2$. Simplified: $\dfrac{2(x-2)}{x-3}$. Height of the hole: $\dfrac{2(-4)}{-5} = \dfrac{8}{5}$. Hole at $\left(-2, \frac{8}{5}\right)$.
- **Zero:** the remaining top factor, $x = 2$.
- **Vertical asymptote:** the remaining bottom factor, $x = 3$.
- **Horizontal asymptote:** degrees equal (2 and 2), so the ratio of leading coefficients: $y = \frac{2}{1} = 2$.
- **y-intercept:** $r(0) = \dfrac{-8}{-6} = \dfrac{4}{3}$.

11. $r(x) = \dfrac{x^2 + 1}{x - 1}$.

- Vertical asymptote: $x = 1$ (the top is $2 \neq 0$ there).
- The top's degree is one more than the bottom's, so there's a slant asymptote. Long division:
  - $x^2 \div x = x$. Multiply: $x(x-1) = x^2 - x$. Subtract: $(x^2 + 0x + 1) - (x^2 - x) = x + 1$.
  - $x \div x = 1$. Multiply: $1(x-1) = x - 1$. Subtract: $(x+1) - (x-1) = 2$.
  - So $r(x) = x + 1 + \dfrac{2}{x-1}$.
- As $x \to \pm\infty$, $\frac{2}{x-1} \to 0$, so the slant asymptote is $\boxed{y = x + 1}$.

12. End-behavior limits.

- $\dfrac{3x+1}{x^2+2}$: divide every term by x^2 to get $\dfrac{\frac{3}{x} + \frac{1}{x^2}}{1 + \frac{2}{x^2}} \to \dfrac{0}{1} = \boxed{0}$ (bottom degree bigger).
- $\dfrac{6x^2 - x}{2x^2 + 5}$: divide by x^2 to get $\dfrac{6 - \frac{1}{x}}{2 + \frac{5}{x^2}} \to \dfrac{6}{2} = \boxed{3}$ (equal degrees).

13. Solve $\dfrac{x-2}{x+1} \geq 0$.

- Critical points: top $= 0$ at $x = 2$; bottom $= 0$ at $x = -1$.
- Test each interval:

Interval	Test x	$\frac{x-2}{x+1}$	Sign
$x < -1$	-2	$\frac{-4}{-1} = 4$	$+$
$-1 < x < 2$	0	$\frac{-2}{1} = -2$	$-$
$x > 2$	3	$\frac{1}{4}$	$+$

- Include $x = 2$ (the expression equals 0 there, and $\geq$ allows it). Exclude $x = -1$ (division by 0).
- $\boxed{x < -1 \text{ or } x \geq 2}$

Equivalent forms

14. $(2x^3 + 3x^2 - x + 5) \div (x + 2)$.

- $2x^3 \div x = 2x^2$. Multiply: $2x^2(x + 2) = 2x^3 + 4x^2$. Subtract: $3x^2 - 4x^2 = -x^2$. Bring down: $-x^2 - x$.
- $-x^2 \div x = -x$. Multiply: $-x(x + 2) = -x^2 - 2x$. Subtract: $-x - (-2x) = x$. Bring down: $x + 5$.
- $x \div x = 1$. Multiply: $1(x + 2) = x + 2$. Subtract: $5 - 2 = 3$.
- $\boxed{2x^2 - x + 1 + \dfrac{3}{x+2}}$
- Check with synthetic division by -2: coefficients 2, 3, −1, 5 give 2, −1, 1 with remainder 3 ✓.

15. Expand $(x + 2)^4$.

- Pascal's row 4: 1, 4, 6, 4, 1. Powers of x count down from 4, and powers of 2 count up from 0.
- $1x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 1(16)$
- $\boxed{x^4 + 8x^3 + 24x^2 + 32x + 16}$

Transformations

W6. $g(x) = f(x - 3) + 2$.

- $x - 3$ inside: shift **right 3**. The $+2$ outside: shift **up 2**.

16. $g(x) = -2f(x + 1) - 4$ with $f(x) = x^2$.

- Inside $x + 1$: shift **left 1**.
- Times 2: **vertical stretch** by 2.
- Negative sign: **reflect over the x-axis**.
- -4: shift **down 4**.
- The vertex of x^2 is $(0, 0)$. Left 1 and down 4 move it to $\boxed{(-1, -4)}$. The reflection makes the parabola open down.

17. $(4, 6)$ is on f. Find a point on $g(x) = 3f(2x) - 1$.

- We know $f(4) = 6$, so we need the input to f to be 4: $2x = 4$, so $x = 2$.
- Then $g(2) = 3f(4) - 1 = 3(6) - 1 = 17$.
- Point: $\boxed{(2, 17)}$
- Pattern: the inside "×2" **halves** x-coordinates; the outside "×3, −1" acts directly on y.

Modeling

18. Quadratic $f(x) = ax^2 + bx + c$ through $(0, 5)$, $(1, 8)$, $(2, 15)$.

- $(0, 5)$: $c = 5$.
- $(1, 8)$: $a + b + 5 = 8$, so $a + b = 3$.
- $(2, 15)$: $4a + 2b + 5 = 15$, so $4a + 2b = 10$, which is $2a + b = 5$.
- Subtract the two equations: $(2a + b) - (a + b) = 5 - 3$, so $a = 2$. Then $b = 1$.
- $f(x) = 2x^2 + x + 5$, and $f(5) = 50 + 5 + 5 = \boxed{60}$.